

## Why are singular values always non-negative?

Asked 8 years ago Modified 10 months ago Viewed 23k times

▲ I have read that the singular values of any matrix  $A$  are non-negative (e.g. [wikipedia](#)). Is there a reason why?

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The first possible step to get the SVD of a matrix  $A$  is to compute  $A^T A$ . Then the singular values are the square root of the eigenvalues of  $A^T A$ . The matrix  $A^T A$  is a symmetric matrix for sure. The eigenvalues of symmetric matrices are always real. But why are the eigenvalues (or the singular values) in this case always non-negative as well?



svd symmetric-matrices



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asked Dec 15, 2016 at 23:25



Louis

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### 4 Answers

Sorted by: Highest score (default) ▾



Assume that  $A$  is real for simplicity. The set of (orthogonal, diagonal, orthogonal) matrices  $(U, \Sigma, V)$  such that  $A = U \Sigma V^T$  is **not a singleton**. Indeed, if  $A = U \Sigma V^T$  then also

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$$A = (-U)(-\Sigma)V^T = U(-\Sigma)(-V^T) = (UD_1)(D_1 \Sigma D_2)(VD_2)^T$$



for any diagonal matrices  $D_1$  and  $D_2$  with only 1 or  $-1$  on the diagonal. Therefore, the positivity of the singular values is **purely conventional**.



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edited Feb 7 at 17:57

answered Oct 20, 2019 at 16:42



Roberto Rastapopoulos

2,212

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I'm assuming that the matrix  $A$  has real entries, or else you should be considering  $A^* A$  instead.

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If  $A$  has real entries then  $A^T A$  is positive semidefinite, since

$$\langle A^T A v, v \rangle = \langle A v, A v \rangle \geq 0$$



for all  $v$ . Therefore the eigenvalues of  $A^T A$  are non-negative.



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answered Dec 15, 2016 at 23:30



carmichael561

54.5k

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11 ▲ This shows that the eigenvalues of  $A^T A$  are non-negative, what about the singular values of  $A$ ? Do you know if the fact that the singular values of  $A$  are (non-negative) square roots of the eigenvalues of  $A^H A$  holds for  $A$  with complex entries? – Learn\_and\_Share Oct 25, 2017 at 8:42

▲ Yes, the singular values of  $A$  are the square roots of the eigenvalues of  $A^* A$ . – carmichael561 Oct 25, 2017 at 15:11

7 ▲ @MedNait Why are they chosen to be the positive square roots? – Undertherainbow Feb 19, 2018 at 15:25

6 ▲ @Undertherainbow I think I read somewhere that it's a convention since, technically, you could also choose the singular values of  $A$  as the negative square roots of the eigenvalues of  $A^H A$ . – Learn\_and\_Share Feb 19, 2018 at 20:50



I think your question is very interesting. Let us take some, non zero singular value  $\sigma_i$ . We can reverse the sign if it is positive. That is,  $-\sigma_i = -\sqrt{\lambda_i^2} = -\lambda_i$  where  $\lambda_i^2$  is an eigenvalue of  $A^T A$  corresponding to an eigenvector  $v_i$ . That is  $A^T A v_i = \lambda_i^2 v_i$ . Who can stop us to write instead  $A^T A(-v_i) = \lambda_i^2(-v_i)$ ? What this means is that we can reverse the sign of a singular value, but then we need to go to the matrix  $V$  and reverse the sign of its corresponding eigenvector column.



Hence, there is not a unique way to write  $A = U \Sigma V^T$ . But if we decide that all  $\sigma_i$  are non-negative, then "yes" there is a unique way to write  $A = U \Sigma V^T$ . Of course all  $\sigma_i$  are sorted from largest to smallest (otherwise there would be a bunch of possibilities by permuting any two columns of  $U$  and  $V$  and their corresponding eigenvalues.)



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edited Jul 29, 2019 at 22:29

answered Jul 29, 2019 at 22:05



Herman Jaramillo

3,162

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Suppose  $T \in \mathcal{L}(V)$ , i.e.,  $T$  is a linear operator on the vector space  $V$ . Then the singular values of  $T$  are the eigenvalues of the positive operator  $\sqrt{T^* T}$ . The eigenvalues of a positive operator are non-negative.

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• Why is  $\sqrt{T^* T}$  a positive operator? Consider  $S = T^* T$ . Then  $S^* = (T^* T)^* = T^* (T^*)^* = T^* T = S$ , and hence  $S$  is self-adjoint. Also,  $\langle S v, v \rangle = \langle T^* T v, v \rangle = \langle T v, T v \rangle \geq 0$  for every  $v \in V$ . Hence  $S$  is positive. Now every positive operator has a unique positive square root, which, for  $S$ , I am denoting with  $\sqrt{T^* T}$ .



• Why are the eigenvalues of a positive operator non-negative? If  $S$  is a positive operator, then  $0 \leq \langle S v, v \rangle = \langle \lambda v, v \rangle = \lambda \langle v, v \rangle$ , and thus  $\lambda$  is non-negative.



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answered Apr 21, 2019 at 6:35



Aditya

891

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